PRODUCER CONSUMER - SIVARATHINAM S (241401102)

Activities Terminal

sivarathinam@localhost~/pc — /usr/bin/vim pc.c

```
ude <stdio.h>
ude <pthread.h>
ude <semaphore.h>

ne BUFFER_SIZE 5

uffer[BUFFER_SIZE];
n = 0;
ut = 0;

 empty;
 full;
 mutex;

*producer(void *arg)

nt item;
or(int i = 0; i < BUFFER_SIZE; i++)

   item = i;
   sem_wait(&empty);
   sem_wait(&mutex);

   buffer[in] = item;
   printf("Produced: %d\n", item);
   in = (in + 1) % BUFFER_SIZE;

   sem_post(&mutex);
   sem_post(&full);


*consumer(void *arg)

nt item;

or(int i = 0; i < BUFFER_SIZE; i++)

   sem_wait(&full);
   sem_wait(&mutex);
```

1,1 Top

Activities Terminal

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```
*consumer(void *arg)

nt item;

or(int i = 0; i <BUFFER_SIZE; i++)

   sem_wait(&ffill);
   sem_wait(&mutex);

   item = buffer[out];
   printf("Consumed: %d\n", item);
   out = (out + 1) % BUFFER_SIZE;

   sem_post(&mutex);
   sem_post(&empty);



main ()

thread_t p, c;

em_init(&empty, 0, BUFFER_SIZE);
em_init(&full, 0, 0);
em_init(&mutex, 0, 1);

thread_create(&p, NULL, producer, NULL);
thread_create(&c, NULL, consumer, NULL);

thread_join(p, NULL);
thread_join(c, NULL);

em_destroy(&empty);
em_destroy(&full);
em_destroy(&mutex);

eturn 0;
```

70,1 Bot

Output :

```
Activities   Terminal                    Mar 24 11:13
                          sivarathinam@localhost:~/pc
oduced: 56
nsumed: 56
oduced: 11

aleb@localhost pc]$ vi pc.c
aleb@localhost pc]$ gcc pc.c
aleb@localhost pc]$ ./a.out pc.c
oduced: 0
nsumed: 0
osumed: 1
oduced: 1
oduced: 2
nsumed: 2
oduced: 3
nsumed: 3
odumed: 4
nsumed: 4
aleb@localhost pc]$ gcc pc.c
aleb@localhost pc]$ ./a.out
oduced: 0
oduced: 0
oduced: 2
oduced: 3
nsumed: 4
nsumed: 0
nsumed: 0
nsumed: 2
nsumed: 3
nsumed: 4
aleb@localhost pc]$ gcc pc.c
aleb@localhost pc]$ ./a.out
aleb@localhost pc]$ ./a.out pc.c
oduced: 0
oduced: 1
oduced: 2
oduced: 3
nsuced: 4
nsumed: 2
nsumed: 3
nsumed: 4
aleb@localhost pc]$
```